

Introduction

AspeQt-2k26 is a modernized version of the classic AspeQt emulator, updated to use the Qt6 framework. It transforms your modern PC into a powerful peripheral server for Atari 8-bit computers (400/800/XL/XE).

Using a standard SIO2PC cable (USB or Serial), AspeQt can emulate:

- **Disk Drives (D1-D8)** - Supports ATR, XFD, PRO images, and direct Folder Mounting.
- **TNFS Client** - Mount and stream disk images directly from the internet (FujiniNet protocol).
- **Modem Bridge** - Connect physical Atari modems to Telnet BBSs via the PC.
- **Printers (P)** - Captures printed output to text or graphical view.
- **Clipboard Bridge (Y)** - Bi-directional copy/paste between PC and Atari.
- **Web Pipe (W)** - Multi-channel gateway for Python scripts and AI.

High-Speed SIO

AspeQt supports "divisor 0" high-speed SIO (approx. 126kbps) with custom OS ROMs (like QMEG or Hiassoft) and standard high-speed protocols (UltraSpeed, Happy, etc.).

Note: Reliable high-speed operation depends on your SIO2PC hardware. FTDI-based USB cables generally offer the best performance.

Disk & Folder Usage

Mounting Disk Images

Drag and drop `.atr`, `.xfd`, or `.pro` files directly into the drive slots (D1-D8). Double-clicking a slot opens the file browser.

Virtual DOS (Folder Images)

You can mount a local PC directory as a writable disk. To boot from a folder, right-click the slot and select **Mount Folder**.

TNFS Client

AspeQt-2k26 includes native support for TNFS (The Network File System), allowing you to mount and stream disk images directly from the internet using the FujiniNet protocol. Access this via the **Tools** menu.

Modem Bridge (Internet Adapter)

The Modem Bridge allows your Atari 8-bit computer to connect to modern Telnet BBSs using your PC's internet connection. It acts as a gateway between your physical Atan hardware and the TCP/IP network.

1. Required Hardware

This feature requires a physical serial interface. It does **not** work with a standard SIO2PC cable alone.

- **Atari Interface:** Atari 850, ICD R-Verter, or P-R Connection.
- **Cables:** A Null Modem cable connecting the Atari Interface to your PC.
- **PC Hardware:** A USB-to-Serial adapter (FTDI recommended).

Setup Tip: Your Null Modem cable **MUST** support the DTR (Data Terminal Ready) signal. AspeQt asserts DTR High to tell the Atari 850 that the "Modem" is powered on. If your cable lacks this wire, the Atari will not respond.

2. Configuration Reference

Configure the bridge settings via **Tools > Options > Modem Bridge**. Correct settings are vital for stable communication.

Setting	Description & Recommended Value
Enable Modem Bridge	Master switch for the feature. When checked, AspeQt opens the serial port and asserts DTR/RTS signals. <i>Note: If unchecked, the Atari will see the modem as "Offline" or "Not Ready".</i>
Serial Port	Select the COM port (Windows) or TTY device (Linux/Mac) corresponding to your USB-to-Serial adapter . Do NOT select your SIO2PC adapter here.
Baud Rate	The communication speed between PC and Atari. Must match your terminal software settings. Recommended: 19200 (Standard for Atari 850).
Flow Control	Enables hardware RTS/CTS handshaking. Recommended: OFF (Unchecked) unless you have a high-quality 5-wire Null Modem cable. If enabled with a 3-wire cable, the connection will freeze.
Local Echo	If checked, AspeQt echoes all typed characters back to the Atari. Use this if you cannot see what you are typing in your terminal software. Recommended: OFF (Most BBS software handles echoing remotely).
Phonebook File	Path to the phonebook.xml file storing your BBS list. Defaults to the application directory.

3. The Phonebook

Access via **Tools > Phonebook**. Use this to store your favorite BBS addresses and credentials.

- **Dialing:** Select an entry and click **Dial**. AspeQt will look up the credentials and connect.
- **Auto-Login:** If you save a Username and Password in the phonebook, you can use the **Macros** listed below to auto-type them.

Getting a Phonebook: You can download a compatible XML or SyncTerm phonebook list from the Telnet BBS Guide: <https://telnetbbsguide.com/lists/new-bbs-systems/>

4. Supported Commands & Macros

The bridge emulates a standard Hayes modem. It supports the following commands from your Atari terminal software (BobTerm, Ice-T, etc.):

Command / Key	Description
<code>ATDT <Name></code>	Dial by Name. Looks up the entry in the Phonebook (e.g., <code>ATDT Level29</code>). This is required to load credentials for macros.
<code>ATDT <IP:Port></code>	Raw Dial. Connects directly to an IP (e.g., <code>ATDT 192.168.1.5:23</code>). Macros are disabled in this mode.
<code>ATH</code>	Hangup. Disconnects the current session.
<code>+++</code>	Escape Sequence. Pauses the data stream and switches the modem to "Command Mode" without disconnecting. (Requires 1 second of silence before and after).
<code>[ESC] + [U]</code>	Type Username. (Macro) Auto-types the Login ID saved in the current Phonebook entry.
<code>[ESC] + [P]</code>	Type Password. (Macro) Auto-types the Password saved in the current Phonebook entry.
<code>[ESC] + [H]</code>	Panic Hangup. Immediately terminates the connection and returns to Command Mode. Use this if the session freezes.

5. ATASCII Translation

Modern internet servers expect standard ASCII input. AspeQt automatically performs the following translations:

- **Backspace:** Atari Delete (Code 126/127) is automatically converted to ASCII Backspace (Code 8).
- **Newlines:** Atari EOL (155) is converted to standard Network Newlines (CR/LF).

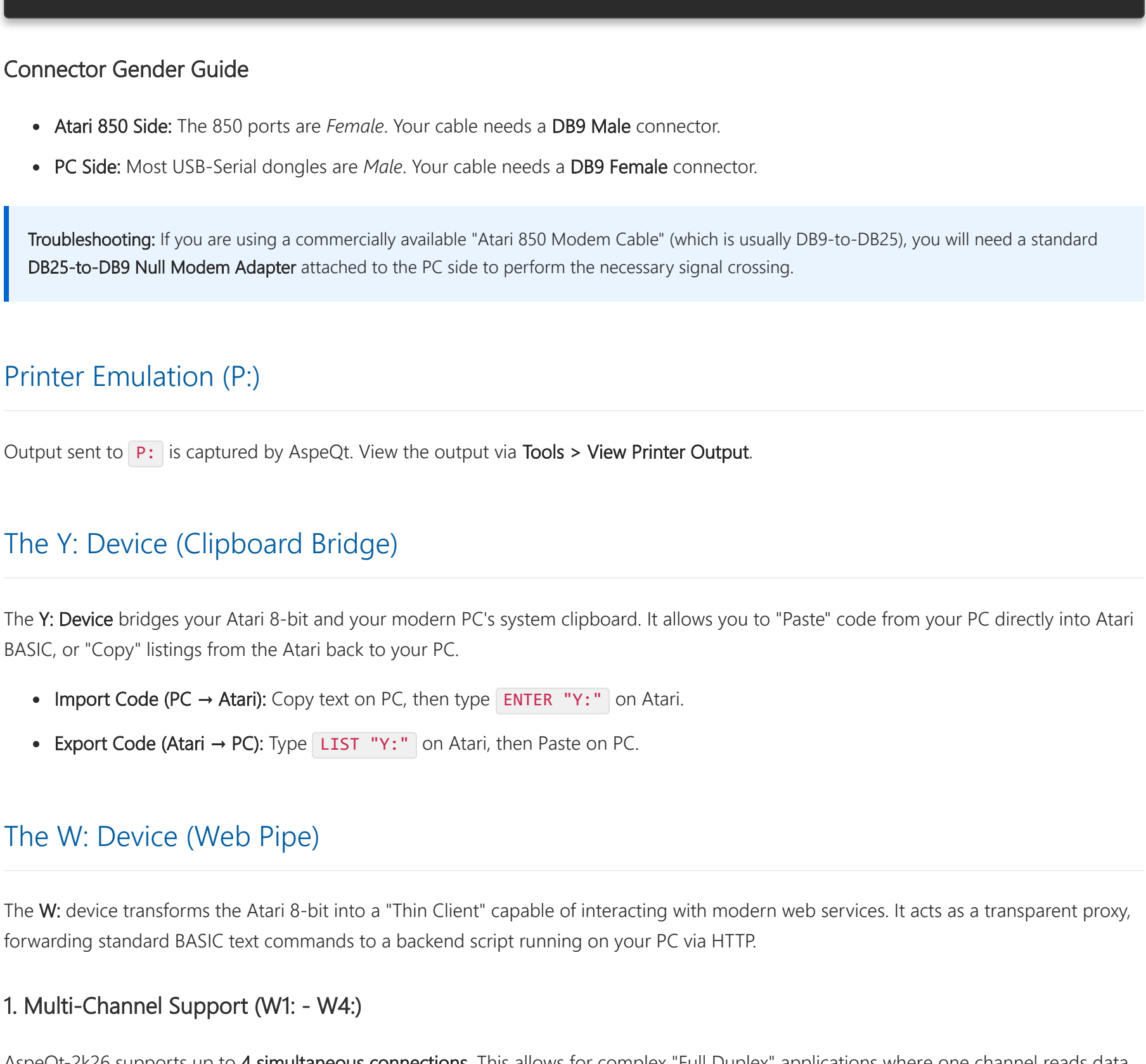
Appendix: Wiring the Atari 850 Null Modem Cable

Unlike modern PCs, the Atari 850 Interface uses a **non-standard DB9 pinout**. You cannot use a standard off-the-shelf "Serial Cable" or "Null Modem Cable" without an adapter, as the wire mappings will be incorrect.

To connect your Atari 850 (Port 1) to a PC USB-Serial Adapter, you must build a custom cable or use a breakout box with the following wiring scheme.

The "Magic" Signal: DTR

The Atari 850 requires the **DSR (Data Set Ready)** and **CD (Carrier Detect)** pins to be HIGH to detect that a modem is present. AspeQt emulates this by asserting its **DTR (Data Terminal Ready)** signal. Your cable **MUST** connect the PC's DTR pin to the Atari's DSR and CD pins, or the bridge will remain "Dead."



Connector Gender Guide

- **Atari 850 Side:** The 850 ports are *Female*. Your cable needs a **DB9 Male** connector.
- **PC Side:** Most USB-Serial dongles are *Male*. Your cable needs a **DB9 Female** connector.

Troubleshooting: If you are using a commercially available "Atari 850 Modem Cable" (which is usually DB9-to-DB25), you will need a standard **DB25-to-DB9 Null Modem Adapter** attached to the PC side to perform the necessary signal crossing.

Printer Emulation (P)

Output sent to `[P]` is captured by AspeQt. View the output via **Tools > View Printer Output**.

The Y: Device (Clipboard Bridge)

The **Y: Device** bridges your Atari 8-bit and your modern PC's system clipboard. It allows you to "Paste" code from your PC directly into Atari BASIC, or "Copy" listings from the Atari back to your PC.

- **Import Code (PC → Atari):** Copy text on PC, then type `[ENTER "Y:"]` on Atari.
- **Export Code (Atari → PC):** Type `[COPY "Y:"]` on Atari, then Paste on PC.

The W: Device (Web Pipe)

The **W:** device transforms the Atari 8-bit into a "Thin Client" capable of interacting with modern web services. It acts as a transparent proxy, forwarding standard BASIC text commands to a backend script running on your PC via HTTP.

1. Multi-Channel Support (W1 - W4)

AspeQt-2k26 supports up to **4 simultaneous connections**. This allows for complex "Full Duplex" applications where one channel reads data while another writes data, or background threads poll for status updates.

- **W:** or **W1:** - Primary Channel (Device \$57)
- **W2:** - Secondary Channel (Device \$56)
- **W3:** - Tertiary Channel (Device \$55)
- **W4:** - Quaternary Channel (Device \$54)

```
10 REM -- QUAD CHANNEL EXAMPLE --
20 OPEN #1,8,8,"W1:http://192.168.1.5:5000/cmd" : REM Command Channel1
30 OPEN #2,4,8,"W2:http://192.168.1.5:5000/data" : REM Data Stream
40 OPEN #3,4,8,"W3:http://192.168.1.5:5000/status":REM Background Poll
50 PRINT #1,"GET STATUS"
60 INPUT #3,STATUS$
70 IF STATUS$="READY" THEN INPUT #2,DATAS
80 CLOSE #1 : CLOSE #2 : CLOSE #3
```

2. Configuration: Text vs. Binary Mode

The **W:** device can be configured globally via **Options > Emulation**, or overridden per-connection using the command parameters.

- **Text Mode:** Best for BASIC and Web APIs.
 - `Write:` Converts Atari EOL (`$9B`) → Unix Newline (`\n`).
 - `Read:` Converts Unix Newline (`\n`) → Atari EOL (`$9B`).

- **Binary Mode:** Best for Game Loaders and Asset Streaming.
 - Data is passed raw, byte-for-byte. No translation occurs.

On-the-Fly Data Switching (Advanced)

You can override the global setting for a specific connection by using the **AUX2** parameter (the 3rd number in the OPEN or XIO command).

- `0:` Use Global AspeQt Settings (Default)
- `1:` Force TEXT Mode (Translate EOLs)
- `2:` Force BINARY Mode (Raw Data)

```
10 REM -- MODE SWITCHING EXAMPLES --
20 US="W:http://192.168.1.5:5000/api"

30 REM Force TEXT Mode (AUX2=1) for API calls
40 OPEN #1, 8, 1, US
50 PRINT #1,"HELLO" : REM sends "HELLO\n"
60 CLOSE #1

70 REM Force BINARY Mode (AUX2=2) for Game Asset.
80 OPEN #1, 8, 2, "W:http://server.com/sprite.dat"
90 REM Data read here will NOT have $9B converted
100 CLOSE #1
```

3. Advanced: XIO Commands (One-Shot)

The **XIO** command allows you to send a "Fire and Forget" packet without the overhead of OPEN/PRINT/CLOSE. This is ideal for telemetry, logging, or high-speed triggers.

Command 80 (\$50): Fast POST

Sends a payload to a URL in a single operation. The format is `"W:URL,DAT"`. It supports all channels (W1-W4) and the AUX2 mode flag.

```
10 DIM PKTS(120)
20 URL$="http://192.168.1.5:5000/score"
30 SCORE=15000

40 REM -- Build Packet --
50 PKTS="W1:" PKTS(4)=URL$
60 PKTS(LEN(PKTS)+1)="," PKTS(LEN(PKTS)+1)+STR$(SCORE)

70 REM -- Execute XIO 80 with FORCE BINARY (AUX2=2) --
80 XIO 80,#1,8,2,PKTS
```

4. Application Examples Library

This library contains a collection of "Hybrid Computing" examples that demonstrate the power of the **W:** Device.

1. Gemini AI Chatbot

Turns the Atari into a client for Google's Gemini Large Language Model. You can type natural language questions ("Who is Captain Kirk?") and receive intelligent, summarized answers on the Atari screen.

- **BASIC File:** GEMINI_AI.LST
- **Python Script:** gemini_ai.py
- **Dependencies:** `pip install google-generativeai`

2. Basic Graphical Weather Station

Fetches live weather data from the Open-Meteo API for a specific location and displays it on the Atari using GRAPHICS 5 (4-color mode).

- **BASIC File:** WEATHER.LST
- **Python Script:** weather.py
- **Dependencies:** `pip install requests`

3. Cloud Notepad

Demonstrates Reading and Writing to the PC's hard drive. It acts as a "Cloud Drive" for text.

- **BASIC File:** CLOUD_NOTE.LST
- **Python Script:** cloud_note.py

4. Text-to-Speech Synthesizer

The "Talking Atari." Any text typed on the Atari is sent to the PC, where it is read aloud using the Linux **espeak** engine.

- **BASIC File:** TTS_SERVER.LST
- **Python Script:** tts_server.py
- **Dependencies:** `sudo apt install espeak` (Linux)

5. Web News Scraper

Connects to a modern news website, scrapes the HTML, strips out ads/javascript, and delivers a plain-text headline summary to the Atari.

- **BASIC File:** SCRAPER.LST
- **Python Script:** scraper.py
- **Dependencies:** `pip install requests beautifulsoup4`

6. Remote Execution (Linux & Windows)

Allows the Atari to execute shell commands or launch GUI applications on the Host PC.

- **BASIC Files:** EXEC_LINUX.LST / EXEC_WIN.LST
- **Python Scripts:** exec_linux.py / exec_win.py

7. File Upload / Telemetry

A utility to upload raw data or text files from the Atari to the PC.

- **BASIC File:** FILE_UPLOAD.LST
- **Python Script:** file_upload.py

Quick Dependency Install

```
# Linux/Mac
sudo apt install espeak
pip install flask requests beautifulsoup4 google-generativeai psutil
```

```
# Windows
pip install flask requests beautifulsoup4 google-generativeai psutil
```